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॥वसुधैव कुटुम्बकम्॥

Name:- Bhoomi Rai

Branch:-CSE

Division:- C3

Prn:- 26070122513

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WINDOWS OS



Introduction

Windows Operating System is a system software developed by Microsoft. It is an operating system that provides an interface between the user and the computer hardware. An operating system is responsible for controlling and managing all the major resources of a computer system, such as the processor, memory, files, storage devices, input devices, output devices, and network connections.

Windows is mainly known for providing a Graphical User Interface (GUI). This means that users can interact with the computer using graphical elements such as windows, icons, menus, buttons, taskbars, and a mouse instead of entering commands for every operation. Because of its simple and user-friendly interface, Windows is widely used in homes, schools, colleges, offices, businesses, and other organizations.

When a user performs an operation such as opening a file, running a program, printing a document, or connecting to the Internet, Windows works in the background to manage the required hardware and software resources. Therefore, Windows acts as an important link between the user, application programs, and computer hardware.

The History and Evolution of Windows

Beginning of Windows: Before the development of Windows, Microsoft primarily provided MS-DOS, which was a command-line operating system. In MS-DOS, the user had to type commands to perform operations such as creating directories, copying files, deleting files, and running programs. Although command-line systems were powerful, they were difficult for beginners because the user needed to remember different commands and their syntax. Microsoft therefore started developing a graphical environment that would make personal computers easier to operate. This graphical environment became Microsoft Windows. The early versions of Windows worked closely with MS-DOS, while later versions gradually developed into complete operating systems based on the Windows NT architecture.

Windows 1.0 — 1985: Windows 1.0 was introduced by Microsoft in 1985. It was the first major version of the Windows graphical environment. Windows 1.0 was not a completely independent operating system in the same sense as modern Windows because it operated on top of MS-DOS. It provided a graphical environment where users could interact with applications through windows, menus, icons, and a mouse. It included basic applications such as Calculator, Paint, Notepad, and other simple utilities. The purpose of Windows 1.0 was to make computer operation easier by reducing dependence on typed commands. Although Windows 1.0 had limited

capabilities compared with modern operating systems, it established the basic idea of a graphical Windows environment and provided the foundation for later versions.

Windows 2.0 — 1987: Windows 2.0 was released in 1987 and introduced several improvements over Windows 1.0. It provided better graphical capabilities and allowed windows to overlap, making it easier for users to work with multiple applications. It also introduced improved keyboard shortcuts and better support for applications. The improved interface made Windows more flexible and useful for personal computer users. Although Windows 2.0 was still dependent on MS-DOS, it represented an important step in the development of Microsoft's graphical operating environment.

Windows 3.0 — 1990: Windows 3.0 was released in 1990 and became much more successful than the earlier versions. It provided a significantly improved graphical interface and better memory management. It introduced important components such as Program Manager and File Manager. Program Manager allowed users to organize and launch applications, while File Manager provided a graphical method for managing files and directories. Windows 3.0 also provided improved graphics and support for a larger range of applications. Its improved performance and graphical capabilities helped Windows become a popular platform for personal computers.

Windows 3.1 — 1992: Windows 3.1 was released in 1992 as an improved version of Windows 3.0. It provided a more stable and user-friendly graphical environment and became one of the most widely used Windows versions of its time. One of its important improvements was support for TrueType fonts, which improved the appearance of text on computer screens and printed documents. Windows 3.1 also provided better multimedia support for sound, video, and animation. It improved application compatibility, memory management, and overall usability. Windows 3.1 played an important role in establishing Windows as a popular graphical platform for personal computers.

Windows 95 — 1995: Windows 95 was one of the most important milestones in the history of Windows. It introduced a completely improved desktop experience and several features that became standard parts of Windows for many years. The Start menu, taskbar, desktop icons, and Recycle Bin became important elements of the Windows interface. Windows 95 also introduced improved Plug and Play technology, which made it easier for users to install and configure hardware devices. It provided better support for 32-bit applications and introduced support for long file names. The Start menu allowed users to access programs, documents, settings, and system functions easily. The taskbar made it easier to switch between running applications. Windows 95 therefore made personal computers considerably easier to use and played a major role in making Windows popular among general users.

Windows 98 — 1998: Windows 98 was released in 1998 and improved many features introduced in Windows 95. It provided better support for hardware devices and Internet technologies. Support for USB devices was improved, making it easier to connect peripherals such as printers, cameras, and storage devices. Windows 98 also improved multimedia capabilities, file management, networking, and system utilities. The operating system was designed for home and personal computer users and became another popular version of Windows. It represented Microsoft's continuing effort to improve hardware compatibility and Internet connectivity.

Windows 2000: Windows 2000 was based on Microsoft's Windows NT architecture and was mainly designed for professional and business environments. Unlike the older consumer versions that were closely connected with MS-DOS, Windows 2000 was built on a more robust architecture designed for stability, security, networking, and professional computing. It provided improved user management, file permissions, networking, hardware support, and system reliability. Windows 2000 was particularly important from an operating-system perspective because the Windows NT architecture later became the foundation for modern mainstream Windows versions.

Windows XP — 2001: Windows XP was released in 2001 and became one of the most successful Windows operating systems. The name XP came from the word "Experience." One of the most important aspects of Windows XP was that it brought Microsoft's consumer Windows experience onto the Windows NT architecture. This provided much better stability and reliability compared with older DOS-based Windows versions. Windows XP introduced a redesigned graphical interface with an improved Start menu, taskbar, icons, and visual appearance. It provided better hardware support, networking, multimedia capabilities, and security. It also included System Restore, which allowed users to restore certain system settings to an earlier state. Windows XP remained popular for many years because of its stability, software compatibility, hardware support, and user-friendly interface.

Windows Vista — 2006: Windows Vista was released in 2006 and introduced major changes in the visual appearance and security of Windows. One of its most noticeable features was the Windows Aero graphical interface, which provided transparency effects, animations, and improved visual effects. Windows Vista also introduced stronger security mechanisms such as User Account Control, which could request confirmation when an operation required elevated privileges. It included improved search functionality, networking features, device management, and security tools. However, Windows Vista was also criticized because it required more hardware resources and had compatibility and performance issues on some systems. These limitations affected its popularity and encouraged Microsoft to focus strongly on performance improvements in the next version.

Windows 7 — 2009: Windows 7 was released in 2009 and became one of the most successful versions of Windows. It was designed to improve many of the performance and usability problems experienced with Windows Vista. Windows 7 provided a refined graphical interface, an improved taskbar, better window management, improved hardware compatibility, and better overall performance. The taskbar allowed users to pin frequently used applications and switch between running programs more conveniently. Windows 7 also introduced Libraries for organizing documents, pictures, music, and videos. It improved networking and provided support for touch-enabled devices. Because of its combination of performance, stability, attractive interface, and compatibility, Windows 7 became extremely popular among personal and business users.

Windows 8 — 2012: Windows 8 was released in 2012 and represented another major change in Windows design. Microsoft designed Windows 8 to work with both traditional computers and touch-screen devices. Instead of the traditional Start menu, Windows 8 introduced a Start screen containing tiles. These tiles could display changing information and were known as Live Tiles. Windows 8 also introduced the Windows Store and provided stronger support for touch interaction. It included improvements in startup performance, networking, security, and system management. However, many traditional desktop users found the new Start screen difficult to use because they were accustomed to the Start menu and keyboard-and-mouse interface. This led Microsoft to make significant changes in the next version.

Windows 8.1 — 2013: Windows 8.1 was released as an improved version of Windows 8. It addressed several user complaints and provided improvements to the Start experience, desktop integration, search, customization, and system performance. Microsoft attempted to provide a better balance between touch-oriented features and the traditional desktop environment. Windows 8.1 continued to support the modern interface introduced in Windows 8 while making the system more convenient for traditional PC users.

Windows 10 — 2015: Windows 10 was released in 2015 and combined traditional Windows features with several modern features introduced in Windows 8. One of the most important changes was the return of the familiar Start menu. The Windows 10 Start menu combined a traditional application list with Live Tiles. Windows 10 introduced features such as virtual desktops, Microsoft Edge, improved Windows Defender, Action Center, and improved Windows Update. Virtual desktops allowed users to organize different applications and activities into separate desktop environments. Windows 10 was designed to work across different types of devices and provided a more unified Windows experience. It also adopted a more continuous update approach, with Microsoft regularly providing security updates and feature improvements.

Windows 11 — 2021: Windows 11 was introduced in 2021 as the next major version of Windows. It provided a redesigned graphical interface with a centered Start menu and redesigned taskbar. Windows 11 placed significant emphasis on productivity, multitasking, security, and modern hardware. One important feature is Snap Layouts, which allows users to arrange multiple application windows into predefined layouts. This makes multitasking easier, especially when working with several applications simultaneously. Windows 11 also introduced redesigned widgets, improvements to the Microsoft Store, updated system applications, and stronger hardware-based security requirements. Technologies such as Secure Boot and TPM became more important in the Windows 11 hardware requirements. The overall design of Windows 11 focuses on a cleaner interface and improved user experience.

Advantages of Windows OS

Windows has several advantages.

Easy to Use

Windows provides a graphical interface that is easy to understand. Users can perform operations using icons, menus, buttons, and a mouse.

Wide Software Availability

A large number of applications are available for Windows, including office software, programming tools, educational applications, design software, multimedia applications, and games.

Hardware Support

Windows supports a large variety of hardware devices such as printers, scanners, webcams, graphics cards, USB devices, and networking devices.

Multitasking

Windows allows multiple applications to run simultaneously.

Gaming Support

Windows is widely used for computer gaming because many PC games are designed specifically for Windows.

Networking

Windows provides support for Internet connections, Wi-Fi, Ethernet, file sharing, printer sharing, and other networking facilities.

User Account Management

Windows allows different users to have separate accounts and permissions on the same computer.

Disadvantages of Windows OS

Although Windows is widely used, it also has some disadvantages.

Security Threats

Because Windows has a large user base, it is frequently targeted by malware and other security threats. Users need to keep the system updated and use appropriate security measures.

Licensing Cost

Some Windows editions require a paid license, which can increase the overall cost of using the operating system.

Resource Requirements

Modern Windows versions can require significant amounts of RAM, storage, and processing power. Older computers may not run newer versions efficiently.

Updates

Windows regularly receives updates. Updates are important for security and improvements, but they may sometimes require system restarts or consume Internet bandwidth.

Less Customization Compared with Some Open-Source Systems

Windows provides customization options, but some open-source operating systems offer greater freedom to modify the operating system.

MAC OS



Introduction:

macOS is a graphical operating system developed by Apple Inc. for Macintosh computers. It provides an interface between the user and computer hardware and manages important resources such as the CPU, memory, storage, files, input and output devices, networking, and security. macOS is known for its graphical user interface, stability, security, multimedia support, and close integration between Apple hardware and software. The history of macOS began with the original Macintosh operating system in 1984 and gradually developed into a modern UNIX-based operating system. Its evolution includes major changes in the graphical interface, system architecture, security, multitasking, memory management, networking, hardware support, and processor architecture.

The History and Evolution of macOS

• The Classic Mac Era (1984–2000):
Apple introduced the Macintosh System Software in 1984 with the original Macintosh computer. It featured one of the first successful graphical user interfaces (GUI), mouse support, and desktop icons. Over time, it evolved into Mac OS 8 and Mac OS 9.

• The OS X Revolution (2001–2012):
In 2001, Apple launched Mac OS X (based on UNIX architecture), bringing improved

stability, memory protection, and multitasking. It introduced features like the Dock, Aqua interface, and Spotlight search.

• **Refinement and Ecosystem Growth (2013–2019):**
Apple rebranded it as macOS in 2016. Versions like macOS Sierra, Mojave, and Catalina enhanced cloud integration, security, and cross-device continuity.

• **Apple Silicon Era (2020–Present):**
With macOS Big Sur and later versions (Monterey, Ventura, Sonoma), Apple transitioned to its own M-series chips, improving performance, battery life, and system efficiency.

Key Features of macOS

Graphical User Interface: macOS provides a graphical user interface that allows users to interact with the computer through windows, icons, menus, buttons, folders, and a mouse or trackpad. The graphical interface makes the system easier to use because users do not need to remember commands for every operation. The desktop, Dock, Finder, menus, and system settings provide a consistent environment for performing different tasks. A user can open applications, manage files, change settings, browse the Internet, and perform other operations through graphical controls.

UNIX-Based Architecture: Modern macOS is built on a UNIX-based foundation and uses the XNU kernel. This provides a strong foundation for process management, memory management, networking, security, and hardware communication. The UNIX foundation also makes macOS useful for programmers and developers because it provides a Terminal environment with many UNIX commands and tools. Therefore, macOS combines a graphical desktop environment with powerful command-line capabilities.

Multitasking: macOS supports multitasking, which means that multiple applications and processes can execute at the same time. For example, a user can open a web browser, create a document, listen to music, and run a programming application simultaneously. The operating system manages the CPU and memory resources of these applications and provides processor time to their processes and threads.

Process Management: macOS manages processes and threads running on the computer. When an application is opened, the operating system creates and manages the required processes. It schedules threads for execution and allocates CPU resources according to system requirements. When an application is closed, macOS releases the resources associated with its processes. Process management helps the system run multiple applications efficiently.

Advantages

• **Highly stable performance:** Since macOS is built specifically for Apple hardware, it runs smoothly with fewer crashes and system errors.

• **Strong security:** Built-in protections like Gatekeeper and System Integrity Protection reduce malware risks.

• **Optimized hardware integration:** Apple designs both hardware and software, resulting in excellent battery life and performance.

• **Professional software support:** Popular creative tools like Final Cut Pro and Logic Pro are optimized for macOS.

• **Seamless Apple ecosystem:** Easy file sharing and syncing between Mac, iPhone, iPad, and Apple Watch.

• **Clean and user-friendly interface:** Minimalistic design makes it easy to navigate.

• **Reliable system updates:** Apple provides long-term updates for supported devices.

Disadvantages

• **Expensive devices:** Macs are generally more costly compared to Windows PCs.

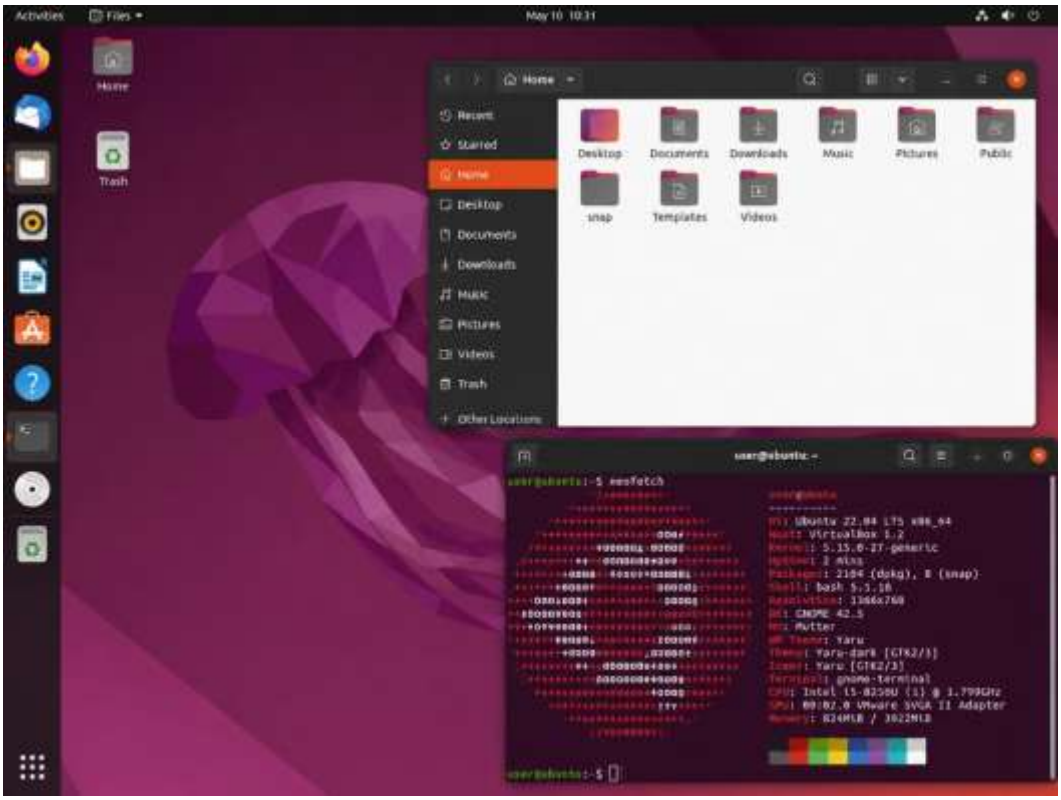
• **Limited hardware options:** Only works on Apple devices, so customization is restricted.

• **Less gaming support:** Fewer games are developed specifically for macOS.

• **Limited upgrade options:** Hardware upgrades are difficult or impossible in newer Mac models.

• **Software compatibility issues:** Some Windows-only applications do not run natively

LINUX OS



Introduction

Linux is an open-source operating system kernel created by Linus Torvalds in 1991. It is free to use, modify, and distribute. Linux powers servers, supercomputers, smartphones (Android), and personal computers. It is known for its stability, security, and flexibility.

The History and Evolution of Linux

- **Birth of Linux (1991):**
Linus Torvalds created the Linux kernel as a free alternative to UNIX. Developers worldwide contributed to its growth.

- **Rise of Distributions (1990s–2000s):**
Linux distributions (distros) like Ubuntu, Red Hat, Debian, and Fedora emerged, making Linux easier to install and use.

- **Enterprise and Server Dominance (2000–2015):**
Linux became the backbone of internet servers, cloud computing, and enterprise systems.

• **Modern Era (2015–Present):**
Linux dominates cloud infrastructure, DevOps environments, embedded systems, and powers Android OS. It is widely used in cybersecurity and programming fields.

Key Features

• **Open-source:** Source code is publicly available and customizable.
• **Multiple distributions:** Ubuntu, Kali Linux, Fedora, Arch Linux, etc.
• **Strong security:** Fewer viruses compared to other OS.
• **Command-line interface (CLI):** Powerful terminal for advanced control.
• **Multi-user support:** Multiple users can work simultaneously.
• **High stability:** Rarely crashes, ideal for servers.
• **Package managers:** Easy software installation via repositories.
• **Customizable interface:** Users can modify themes, kernels, and desktops.
• **Lightweight versions:** Can run on old or low-spec hardware.
• **Server dominance:** Powers most web servers worldwide.

Advantages

• **Free and open-source:** Users can download, modify, and distribute it without cost.

• **Highly secure:** Linux has strong permission controls and fewer viruses.

• **Stable and reliable:** Rarely crashes, making it ideal for servers and development work.

• **Customizable:** Users can modify the interface, kernel, and system behavior.

• **Lightweight versions available:** Can run efficiently on older or low-spec computers.

• **Strong community support:** Large global developer community provides help and updates.

• **Excellent for programming:** Preferred OS for developers and cybersecurity professionals.

Disadvantages

• **Steep learning curve:** Beginners may find it difficult, especially when using terminal commands.

• **Limited commercial software support:** Some popular applications (like certain Adobe tools) are not available.

• **Gaming limitations:** Fewer games compared to Windows.

• **Hardware driver issues:** Some devices may not have proper driver support.

• **Fragmentation:** Too many distributions can confuse new users.

ANDROID OS



Introduction

Android is an open-source mobile operating system developed by Google. It is primarily used in smartphones, tablets, smart TVs, and wearable devices. Android is based on the Linux kernel and is the most widely used mobile OS globally.

The History and Evolution of Android

• **Early Development (2003–2008):**
Android Inc. was founded in 2003 and acquired by Google in 2005. The first Android phone (HTC Dream) was launched in 2008.

• **Dessert Naming Era (2009–2018):**
Android versions were named after desserts like Cupcake, KitKat, Lollipop, Marshmallow, Oreo, and Pie.

• **Modern Android (2019–Present):**
Google shifted to numeric naming (Android 10, 11, 12, 13, 14). Focus areas include privacy, AI features, improved UI design (Material You), and enhanced security.

Key Features

• **Google Play Store:** Large marketplace for apps and games.
• **Customization:** Widgets, launchers, themes, and layouts.
• **Multitasking:** Split-screen and background apps.
• **Open ecosystem:** Used by multiple manufacturers (Samsung, OnePlus, Xiaomi).
• **Google Assistant integration:** Voice-controlled smart features.
• **Notification system:** Advanced and interactive notifications.
• **Security features:** Biometric authentication, encrypted storage.
• **Regular updates:** Security patches and version upgrades.
• **Cloud sync:** Google account integration.
• **Wide hardware support:** Works on phones, TVs, cars, and smart devices.

Advantages

• **Open ecosystem:** Available on devices from many manufacturers, offering wide choices.

• **Highly customizable:** Users can change themes, launchers, widgets, and layouts.

• **Large app availability:** Google Play Store offers millions of applications.

• **Affordable device range:** Available in budget, mid-range, and premium models.

• **Multitasking support:** Supports split-screen and background app operations.

• **Expandable storage:** Many Android phones support SD cards.

• **Wide hardware compatibility:** Runs on smartphones, TVs, tablets, and smart devices.

Disadvantages

• **Security risks:** Open ecosystem can make devices more vulnerable to malware.

• **Fragmented updates:** Not all devices receive updates at the same time.

• **Pre-installed apps (bloatware):** Some manufacturers include unwanted apps.

• **Performance variation:** Quality and performance depend on device manufacturer.

• **Battery optimization issues:** Some apps may consume excessive power.

iOS



Introduction

iOS is a proprietary mobile operating system developed by Apple Inc. It runs exclusively on Apple devices such as the iPhone. It is known for its high security, optimized performance, and tightly controlled ecosystem.

The History and Evolution of iOS

• **Initial Release (2007):**
Launched with the first iPhone in 2007 as “iPhone OS.” It introduced a revolutionary multi-touch interface.

• **App Store Launch (2008):**
The App Store allowed third-party developers to create apps, expanding functionality massively.

• **Growth and Stability (2010–2016):**
Features like Siri, iMessage, FaceTime, and Control Center were introduced.

• **Modern iOS (2017–Present):**
Focus on privacy, AI enhancements, Face ID, widgets, and deeper ecosystem integration with Apple devices.

Key Features

• **App Store:** Secure and regulated application marketplace.
• **Face ID & Touch ID:** Biometric security authentication.
• **Siri:** Intelligent voice assistant.
• **iMessage & FaceTime:** Encrypted communication services.
• **iCloud integration:** Automatic data syncing and backup.
• **Strong privacy controls:** App tracking transparency.
• **Optimized performance:** Designed specifically for Apple hardware.
• **Regular updates:** Long-term device support.
• **AirDrop:** Wireless file sharing between Apple devices.
• **Screen Time:** Monitor and manage device usage.

Advantages

• **High security and privacy:** Strict app review process reduces malware risks.

• **Smooth performance:** Optimized specifically for Apple hardware.

• **Regular updates:** Apple provides updates to devices for many years.

• **Strong ecosystem integration:** Seamless connection with other Apple devices.

• **User-friendly interface:** Simple and consistent design.

• **Quality app optimization:** Apps are well-optimized and stable.

• **Excellent customer support:** Strong service network and support system.

Disadvantages

• **Expensive devices:** iPhones are generally costly.

• **Limited customization:** Users cannot modify interface freely like Android.

• **Restricted file access:** Less flexibility in file management.

• **No expandable storage:** Storage cannot be increased with memory cards.

• **Closed ecosystem:** Works best only within Apple products.

SUMMARY

- An Operating System (OS) is system software that acts as an interface between the user and computer hardware. It manages hardware and software resources and provides an environment in which application programs can run. The operating system performs important functions such as process management, memory management, file management, device management, security, and user interface management. Without an operating system, it would be difficult for users to operate a computer or mobile device efficiently. Examples of operating systems include Windows, macOS, Linux, Android, and iOS.
- Windows OS is an operating system developed by Microsoft and is widely used on personal computers and laptops. It provides a graphical user interface (GUI), making it easy for users to access files, applications, settings, and other system functions. Windows supports a large number of software applications, hardware devices, networking features, and gaming applications. Important features include a Start Menu, Taskbar, File Explorer, Windows Security, multitasking, device support, and regular updates. Different versions of Windows have been released over time, such as Windows 7, Windows 8, Windows 10, and Windows 11. Windows is commonly used in homes, schools, offices, and businesses.
- macOS is an operating system developed by Apple for Mac computers such as MacBook and iMac. It provides a clean and user-friendly graphical interface and is designed to work closely with Apple hardware. macOS includes features such as Finder for file management, Dock for accessing applications, Spotlight Search, strong security, multitasking, and integration with other Apple devices. It is commonly used for education, business, software development, graphic design, video editing, and other professional activities. macOS is known for its smooth performance, security, and integration with the Apple ecosystem.
- Linux is an open-source operating system based on the Linux kernel. Unlike proprietary operating systems, its source code is available for modification and distribution. Linux is available in different distributions, such as Ubuntu, Fedora, Debian, and Linux Mint. It provides both a graphical user interface and a powerful command-line interface called the Terminal. Linux is widely used in

servers, cloud computing, cybersecurity, software development, supercomputers, and personal computers. Its important advantages include security, stability, flexibility, customization, and open-source availability. Ubuntu is one of the popular Linux distributions used for desktop and educational purposes.

- Android OS is a mobile operating system developed primarily for smartphones, tablets, televisions, and other smart devices. It is based on the Linux kernel and provides a touch-based graphical user interface. Android allows users to install applications through the Google Play Store and provides features such as notifications, widgets, multitasking, camera support, Bluetooth, Wi-Fi, GPS, and mobile security. Android is used by many smartphone manufacturers and is one of the most widely used mobile operating systems. Examples of Android devices include smartphones and tablets from different manufacturers.
- iOS is Apple's mobile operating system designed mainly for the iPhone. It provides a touch-based interface and is designed specifically to work with Apple hardware. iOS offers features such as App Store, Face ID or Touch ID, Siri, iCloud, notifications, multitasking, privacy controls, and strong security. It provides integration with other Apple products and services, such as Mac, Apple Watch, and AirPods. iOS is known for its smooth user experience, security, regular software updates, and controlled application ecosystem.

Conclusion

In conclusion, all operating systems perform the basic function of managing hardware and software resources and providing an interface for users. However, they are designed for different purposes. Windows is widely used on personal computers, macOS is designed for Apple Mac computers, Linux is popular for open-source, server, and development environments, Android is mainly used on mobile and smart devices, and iOS is designed for Apple's iPhone. Therefore, the choice of operating system depends on the device, user's requirements, security needs, available applications, performance, and level of customization.